

② Trends and Cloud Services Overview

① Benefits of Serverless Architecture

Serverless architecture allows developers to build and run applications without managing infrastructure. Benefits include:

- Scalability: Automatically scales with demand.
- Cost-Efficiency: Pay only for actual usage, reducing costs.
- Reduced Operational Overhead: No need to manage servers, allowing developers to focus on code.
- Faster Time-to-Market: Accelerates development and deployment cycles.

② Progressive Web Apps (PWAs)

PWAs are web applications that offer native app-like experiences. Key features:

- Offline Access: Functionality without an internet connection.
- Installability: Can be added to the user's home screen.

→ Responsive Design: Adapts to various screen sizes and devices.

→ Push Notifications: Engages users with timely updates.

PWAs enhance user engagement and cost-effective alternatives to native apps.

③ Role of AI and Machine Learning in Software Architecture,

AI and ML are transforming software architecture by:

→ Enhancing User Experience: Personalized recommendations and intelligent interfaces.

→ Automating Processes: Predictive analytics and decision-making support.

Cloud Computing Service Models

→ Infrastructure as a Service (IaaS): Provides virtualized computing resources over the internet.

• Use Case: Flexible and scalable hosting solutions.

→ Platform as a Service (PaaS): offers a platform allowing customers to develop, run and manage applications.

- Use Case: Streamlined application development without managing underlying infrastructure,